Casey Barton

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PROFILE

AI systems engineer based in Honolulu, remote-first. I design, ship, and operate production AI agents — retrieval, reasoning, tool use, evaluation, safety guardrails — in domains where reliability directly affects outcomes.

Three years running a governed agent control plane: multi-step orchestration with error recovery and human-in-the-loop escalation, evaluation frameworks wired into CI/CD (offline benchmarks, regression suites, LLM-as-judge), observability, cost-aware LLM workflows. Approval gates on consequential actions; no completion claim without independent readback.

Earlier career as a home inspector: translating observed evidence into prioritized, actionable recommendations — the same diagnostic discipline I bring to production agent systems.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · webhooks · Supabase · Postgres · Vercel · GitHub Actions (CI/CD) · containers

AI & Agents: LLM APIs · function calling · structured outputs · RAG · agent frameworks · multi-agent orchestration · LLM-as-judge evaluation · MCP protocol · prompt & context engineering

Reliability: eval harnesses · offline benchmarks · regression suites · safety tests · observability · cost-efficient LLM workflows · typed state transitions · human-in-the-loop escalation · fail-closed recovery

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Design, build, and operate production AI agents across a **1,398-repository** platform — full lifecycle from prototype to production: retrieval, reasoning, tool use, evaluation, and safety guardrails.
- Own reliability, observability, and cost efficiency of LLM workflows at scale — typed state transitions, approval gates on consequential actions, recoverable runs; nothing ships without eval validation.
- Run evaluation frameworks wired into CI/CD: offline benchmarks, regression suites, safety tests, and LLM-as-judge pipelines that gate deployment — failure-injection recovery tested before it is needed.
- Multi-step agent orchestration: planning, memory, tool use, error recovery, and human-in-the-loop escalation for runs that degrade gracefully instead of failing silently.
- Build RAG pipelines over my own knowledge estate — retrieval, re-ranking, and context engineering across the memory mesh and estate map, prompt tuning validated by eval harnesses before anything ships.
- Operate a **989-test** agent control plane with **95/95** verification checks green; integrated **65 MCP connectors** plus APIs, webhooks, and data pipelines into one coherent, verifiable system.

EDUCATION

University of Hawaii at Manoa 2016

Bachelor of Science

AWS Cloud Institute 2025

Cloud Computing program

Colorado State University

Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions.
- Recovery restricted to contingency evidence declared before reassessment — the engine never silently tunes values until they pass.

GlacierEQ estate — agent systems portfolio

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Connects multiple technical surfaces — agents, APIs, databases — into one coherent, verifiable working system.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / Agents: LLM APIs · function calling · structured outputs · RAG · agent frameworks · multi-agent orchestration · LLM evaluation · MCP protocol

Reliability: eval harnesses · offline benchmarks · regression suites · LLM-as-judge · observability · verification gates · audit receipts · idempotent automation

Practice: prototype-to-production · human-in-the-loop escalation · error recovery · cost-efficient LLM workflows

Evidence boundary: portfolio repositories are independent work. No healthcare, clinical, or HIPAA experience is claimed; domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.